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Radiographing apparatus

Abstract

An X-ray imaging apparatus includes a sensor panel that converts incident radiation to an electric signal, a tabular metallic base plate including a front surface where the sensor panel is supported and a region where the tabular metallic base plate has a basic thickness, a circuit board disposed on a rear surface of the tabular metallic base plate, the rear surface being opposite to the front surface, wherein a thickness of the tabular metallic base plate in a region where the circuit board is disposed is larger than the basic thickness.

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Background/Summary

BACKGROUND OF THE INVENTION

Field

(1) The present disclosure relates to a radiographing apparatus to be applied, for example, to medical imaging systems.

Description of the Related Art

(2) X-ray imaging apparatuses are widely used, for example, in the field of medicine. An X-ray imaging apparatus emits X-rays to a subject and forms an image based on the intensity distribution of the X-rays that have passed through the subject. A portable X-ray imaging apparatus is available, and such an X-ray imaging apparatus needs to be lightweight for the convenience of portability. The X-ray imaging apparatus is placed under a subject when in use and the weight of the subject is applied thereto during image capturing. Accordingly an increased strength is needed for the X-ray imaging apparatus. The increased strength is needed also because a user may drop the portable X-ray imaging apparatus. Accordingly, weight reduction and increased strength are needed for the portable X-ray imaging apparatus.

(3) Japanese Patent Laid-Open No. 2019-113403 discloses an apparatus that includes a support member supporting a radiation detection panel, cylindrically shaped multiple first projections and a second projection. The first projections and the second projection are formed on a surface of the support member, the surface being opposite to a surface on which the radiation detection panel is supported. The second projection is shorter in length than the first projections in a direction normal to the surface on which the radiation detection panel is supported.

(4) According to Japanese Patent Laid-Open No. 2019-113403, the thickness of the support member is uniform, and the first projections, which are in contact with the inside surface of a housing, are disposed uniformly. Accordingly, the strength is uniform over the entire apparatus. The apparatus having a uniform strength over the entire apparatus, however, does not necessarily lead to weight reduction because, for example, the projections are formed in a region in which the strength is not demanded. In other words, the apparatus cannot necessarily achieve an appropriate balance between weight reduction and increased strength.

SUMMARY

(5) Accordingly, the present invention provides a radiographing apparatus that can achieve an appropriate balance between weight reduction and increased strength.

(6) According to an aspect of the disclosure, a radiographing apparatus includes a sensor panel configured to convert incident radiation to an electric signal, a tabular metallic base plate including a front surface where the sensor panel is supported and a region where the tabular metallic base plate has a basic thickness, and a circuit board disposed on a rear surface of the tabular metallic base plate, the rear surface being opposite to the front surface, wherein, a thickness of the tabular metallic base plate in a region where the circuit board is disposed is larger than the basic thickness.

(7) Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A, 1B, and 1C are views illustrating an X-ray imaging apparatus according to an embodiment.
- (2) FIGS. 2A and 2B are views illustrating the X-ray imaging apparatus according to the embodiment.
- (3) FIGS. 3A and 3B are views illustrating a tabular metallic base plate of the X-ray imaging apparatus according to the embodiment.
- (4) FIGS. 4A, 4B, and 4C are views schematically illustrating the X-ray imaging apparatus when a load is applied to the X-ray imaging apparatus.
- (5) FIGS. 5A and 5B are other views schematically illustrating the X-ray imaging apparatus when a load is applied to the X-ray imaging apparatus.

DESCRIPTION OF THE EMBODIMENTS

- (6) An embodiment of the present disclosure will be described with reference to the attached drawings.
- (7) In the present embodiment, a portable X-ray imaging apparatus, which is otherwise called an “electronic cassette”, will be described as an example of a radiographing apparatus to which the present disclosure is applied.
- (8) FIGS. 1A, 1B, and 1C are views illustrating the X-ray imaging apparatus according to the present embodiment, in which FIG. 1A is a rear view, FIG. 1B is a cross-sectional view taken along line IB-IB, and FIG. 1C is a cross-sectional view taken along line IC-IC. For description purposes, the up-down direction in FIG. 1A is referred to as a “longitudinal direction” of the X-ray imaging apparatus, and the right-left direction in FIG. 1A is referred to as a “transverse direction” of the X-ray imaging apparatus.
- (9) The X-ray imaging apparatus includes a flat and box-like housing **100** that covers the X-ray imaging apparatus. The housing **100** includes a front frame **101**, an entrance face plate **102** attached to the front frame **101**, and a rear housing segment **103**.
- (10) The front frame **101** is made of, for example, a magnesium alloy or an aluminum alloy. X-rays are incident on the front surface of the entrance face plate **102**. The entrance face plate **102** is made of, for example, a lightweight and high-rigidity material having a low X-ray absorption property, such as a carbon fiber reinforced plastic (CFRP). The rear housing segment **103** is made of, for example, a magnesium alloy or an aluminum alloy. The front frame **101** and the rear housing segment **103** are fixed to each other using fastening members **104**, such as screws. A gasket (not illustrated) is provided between the front frame **101** and the rear housing segment **103**, which makes the inside of the housing **100** a sealed space.
- (11) As illustrated in FIG. 1B, a shock absorbing member **105** for absorbing shocks is disposed on the entrance face plate **102** in the sealed space inside the housing **100**. A fluorescent member **106** is disposed behind the shock absorbing member **105**. The fluorescent member **106** is made of a fluorescent material, such as GOS or CsI, which scintillates in response to the incident X-rays that have passed through a subject. A sensor panel **107** is secured to the rear surface of the fluorescent member **106**. The sensor panel **107** can store electric charges when the sensor panel **107** receives light emitted from the fluorescent member **106**. The sensor panel **107** then converts the incident radiation into electric signals. A shield member **108** is disposed on the rear surface of the sensor panel **107** to reduce the impact of the X-rays that have passed through the sensor panel **107**. A tabular metallic base plate **109** that is shaped substantially rectangularly is disposed on the rear surface of the shield member **108**. The tabular metallic base plate **109** is made of, for example, a magnesium alloy or an aluminum alloy. The front surface of the tabular metallic base plate **109** supports the sensor panel **107** and other members. The fluorescent member **106**, the sensor panel

107, the shield member **108**, and the tabular metallic base plate **109** are secured to each other using, for example, a double-sided adhesive tape. The front frame **101**, the rear housing segment **103**, and the tabular metallic base plate **109** can be made of the same metal, such as a magnesium alloy.

(12) Multiple electric circuit boards **110** are disposed on the rear surface of the tabular metallic base plate **109**, the rear surface being opposite to the front surface. The rear surface of the tabular metallic base plate **109** supports the electric circuit boards **110**. The electric circuit boards **110** receive the electric charges stored in the sensor panel **107** and thereby process images. The electric circuit boards **110** also control the X-ray imaging apparatus. Some of the electric circuit boards **110** are connected to the sensor panel **107** via flexible circuit boards **111** through which electric signals are transmitted from the sensor panel **107** to the electric circuit boards **110**. The flexible circuit boards **111** have respective integrated circuits (ICs) **112** that amplify the electric signals from the sensor panel **107**.

(13) As illustrated in FIG. 1C, an opening **103a** that is shaped substantially rectangularly is formed in the rear housing segment **103**. A battery holding member **113** is secured to the inside surface of the rear housing segment **103** using, for example, an adhesive or a double-sided adhesive tape at a position corresponding to the opening **103a**. A battery accommodation space **113a** is formed through the battery holding member **113** to oppose the opening **103a**. The battery accommodation space **113a** is a rectangularly shaped through-hole, and a battery **114** is accommodated and secured therein.

(14) A metallic cover **115**, which is made of aluminum alloy or the like, is detachably attached to the rear housing segment **103** to cover the opening **103a**. A gasket **116** is disposed between the inside surface of the cover **115** and the battery holding member **113** to surround the battery accommodation space **113a**. The gasket **116** is, for example, secured to the inside surface of the cover **115** using an adhesive or a double-sided adhesive tape.

(15) A locking member **117** is fixed to the inside surface of the cover **115** by a screw or the like (not illustrated). The locking member **117** includes an urging spring **118** and a latch **119**. The urging force of the urging spring **118** causes the latch **119** to protrude sideways from the cover **115** and engage the edge of the opening **103a**. When the cover **115** is installed, the urging force of the urging spring **118** causes the latch **119** to engage the edge of the opening **103a** and to bring the cover **115** into a locked state in which the cover **115** is not detachable. In the locked state, the gasket **116** is pressed flat to maintain the sealing of the battery accommodation space **113a**, which can, for example, prevent water from entering the accommodation space for the battery **114** when washing the apparatus. When the cover **115** is removed, the latch **119** is pressed in against the urging force of the urging spring **118**, which unlocks and releases the cover **115** from the edge of the opening **103a**.

(16) Next, components disposed inside the X-ray imaging apparatus of the present embodiment will be described with reference to FIGS. 2A and 2B.

(17) FIGS. 2A and 2B are views illustrating the X-ray imaging apparatus of the present embodiment, where FIG. 2A is a rear view when the rear housing segment **103** is removed and FIG. 2B is a cross-sectional view taken along line IIB-IIB in FIG. 2A.

(18) As described above, the rear surface of the tabular metallic base plate **109** supports multiple electric circuit boards **110**, with some of the electric circuit boards **110** connected to the sensor panel **107** via the flexible circuit boards **111**.

(19) The electric circuit boards **110** can have different shapes depending on their functions. For example, electric circuit boards **110a** to be connected to the sensor panel **107** via the flexible circuit boards **111** are disposed along respective edges of the tabular metallic base plate **109**. Each of the electric circuit boards **110a** has an elongated rectangular shape with a length almost equal to the longitudinal length or the transverse length of the sensor panel **107**. Electric circuit boards **110b** process images and control the X-ray imaging apparatus and have many electric elements, which need to be arranged efficiently. Each of the electric circuit boards **110b** has a rectangular shape with

longer sides exceeding 100 mm, which is approximately twice as long as the short sides. In the present embodiment, two electric circuit boards **110b** are disposed on one side of the rear surface of the X-ray imaging apparatus in the longitudinal direction of the X-ray imaging apparatus (i.e., on the upper side of the X-ray imaging apparatus in FIG. 2A). The two electric circuit boards **110b** are arranged side by side in the transverse direction of the X-ray imaging apparatus.

(20) The ICs **112**, which are installed in respective flexible circuit boards **111**, are readily affected by, for example, vibration. Accordingly, each IC **112** contacts the tabular metallic base plate **109** with an elastic member **120** interposed there between. The elastic member **120** is made of, for example, rubber.

(21) The battery **114** is accommodated in the battery holding member **113** at a position different from where the electric circuit boards **110** are disposed. In other words, the battery **114** is disposed on the other side of the rear surface of the X-ray imaging apparatus in the longitudinal direction of the X-ray imaging apparatus (i.e., on the lower side of the X-ray imaging apparatus in FIG. 2A).

(22) Next, the tabular metallic base plate **109** of the X-ray imaging apparatus according to the present embodiment will be described with reference to FIGS. 3A and 3B, FIGS. 4A to 4C, and FIGS. 5A and 5B.

(23) FIGS. 3A and 3B are views illustrating the tabular metallic base plate **109** of the X-ray imaging apparatus of the present embodiment, where FIG. 3A is a rear view and FIG. 3B is a cross-sectional view taken along line IIIB-IIIB in FIG. 3A.

(24) Ribs **109a** are formed to protrude from the rear surface of the tabular metallic base plate **109**. The ribs **109a** serve as support portions that abut and support the inside surface of the rear housing segment **103** of the housing **100**. In one exemplary embodiment, the ribs **109a** can always be contact the inside surface of the housing **100**. In another exemplary embodiment, the ribs **109a** can oppose the inside surface of the housing **100** with gaps interposed there between. In this embodiment, the ribs **109a** can contact the inside surface of the housing **100** when the tabular metallic base plate **109** deforms due to a load being applied thereto.

(25) In the present embodiment, the ribs **109a** are disposed to surround each of the two electric circuit boards **110b** arranged side by side in the transverse direction of the X-ray imaging apparatus. The ribs **109a** are also disposed to surround each of the three rectangular regions arranged side by side at a position under the two electric circuit boards **110b**.

(26) The rear surface of the tabular metallic base plate **109** are divided into a first region R1, a second region R2, a third region R3, and a fourth region R4 depending on members disposed therein. The first region R1 is a region where a member, such as an electric circuit board **110** or a battery **114**, is not disposed. The first region R1 has a basic thickness $t1$. The second region R2 is a region where the electric circuit boards **110** are disposed. The third region R3 is a region where the ICs **112** installed in respective flexible circuit boards **111** are disposed. The fourth region R4 is a region where the battery holding member **113** is disposed. The battery **114**, which serves as a power supply member, is accommodated and secured in the battery holding member **113**.

(27) The regions R1 to R4 will now be described in more detail.

(28) The first region R1 is the region where a member, such as the electric circuit board **110** or the battery **114**, is not disposed. The first region R1 has the basic thickness $t1$. The first region R1 is larger in area compared with the other regions R2 to R4. The first region R1 can be designed more flexibly because no member is disposed on the rear surface of the tabular metallic base plate **109**, which enables the ribs **109a** to be disposed more flexibly to decrease the distance between adjacent ribs **109a**. In the present embodiment, the distance between adjacent ribs **109a** is smaller in the first region R1 than in the second region R2.

(29) The strength of the apparatus can be obtained by adjusting the thickness of the tabular metallic base plate **109** and by disposing the ribs **109a**. This will be described in more detail with reference to FIGS. 4A to 4C. FIGS. 4A to 4C are views schematically illustrating a state where a load F is applied onto the X-ray imaging apparatus when the X-ray imaging apparatus is placed on a bed **121**

with the front surface facing upward and a subject lies over the X-ray imaging apparatus.

(30) FIGS. 4A and 4B illustrate examples where the distances between adjacent ribs **109a** are large and small, while the thickness of the tabular metallic base plate **109** remains the same.

(31) FIG. 4A is a comparative example in which the distance between adjacent ribs **109a** of the tabular metallic base plate **109** is large. The same reference signs used for the X-ray imaging apparatus of the present embodiment are also used here for ease of description. The load **F** causes the entrance face plate **102** and the shock absorbing member **105** to deform. The deformation of the shock absorbing member **105** applies an additional load to the fluorescent member **106**, the sensor panel **107**, the shield member **108**, and the tabular metallic base plate **109**. In this case, the tabular metallic base plate **109** deforms between adjacent ribs **109a** because of the large distance there between, which can lead to breakage of the sensor panel **107**.

(32) FIG. 4B is an example where the distance between adjacent ribs **109a** of the tabular metallic base plate **109** is made small in the first region **R1**. The load **F** causes the entrance face plate **102** and the shock absorbing member **105** to deform. The deformation of the shock absorbing member **105** applies an additional load to the fluorescent member **106**, the sensor panel **107**, the shield member **108**, and the tabular metallic base plate **109**. In this case, the deformation of the tabular metallic base plate **109** can be suppressed between the ribs **109a** where the distance is small. Even in the case of the thickness **t1** being relatively small in the first region **R1**, the ribs **109a** can be disposed flexibly to increase the rigidity in the first region **R1** where a high rigidity is demanded. As a result, the balance between weight reduction and increased strength can be optimized for the X-ray imaging apparatus.

(33) The second region **R2** is a region where the electric circuit boards **110** are disposed. Due to the electric circuit boards **110** being disposed on the rear surface of the tabular metallic base plate **109**, the design flexibility is limited in the second region **R2**. Because the distance between adjacent ribs **109a** is large and the disposal of the ribs **109a** is less flexible, the second region **R2** is vulnerable to the breakage described in relation to FIG. 4A.

(34) In the second region **R2**, the thickness **t2** of the tabular metallic base plate **109** is made larger than the basic thickness **t1** to obtain rigidity. FIG. 4C illustrates an example in which the distance between adjacent ribs **109a** remains the same as that in FIG. 4A but the thickness of the tabular metallic base plate **109** is made larger compared with the examples in FIGS. 4A and 4B. The load **F** causes the entrance face plate **102** and the shock absorbing member **105** to deform. The deformation of the shock absorbing member **105** applies an additional load to the fluorescent member **106**, the sensor panel **107**, the shield member **108**, and the tabular metallic base plate **109**. In this case, the deformation of the tabular metallic base plate **109** can be suppressed between adjacent ribs **109a** because the increased thickness of the tabular metallic base plate **109** increases the rigidity. Even in the case in which the distance between adjacent ribs **109a** is large and the disposal of the ribs **109a** is less flexible in the second region **R2**, the thickness of the tabular metallic base plate **109** can be increased to increase the rigidity in the second region **R2** where a high rigidity is demanded. Accordingly, the balance between weight reduction and increased strength can be optimized for the X-ray imaging apparatus.

(35) The area of the second region **R2** can be set to be substantially equal to the area in which the electric circuit boards **110** are supported, and the ribs **109a** can be disposed to surround each electric circuit board **110**. Accordingly, a desired strength can be obtained with a minimum increase in the mass.

(36) The thickness of the tabular metallic base plate **109** does not need to be uniform in the second region **R2**. The part having the thickness **t2** can be formed in a grid-like manner in the second region **R2**.

(37) The third region **R3** is a region where the ICs **112** installed in the flexible circuit boards **111** are disposed. In other words, the third region **R3** is a region that overlaps the ICs **112** as viewed in a direction normal to the rear surface of the tabular metallic base plate **109**. The ICs **112** are small

components, and accordingly, the area of the third region R3 is smaller than the area of the second region R2. The ICs 112 are electronic components that affect the image quality considerably even if they are small. The ICs 112 are vulnerable, for example, to vibrations, and the vibrations can cause noises in the image. Accordingly, as illustrated in FIG. 2B, each IC 112 contacts the tabular metallic base plate 109 with an elastic member 120, such as rubber, being interposed there between. The ICs 112 tend to receive external forces readily. Accordingly, the tabular metallic base plate 109 needs to have a strength. It is difficult, however, to provide the ribs 109a in this region because the flexible circuit boards 111 are present.

(38) In order to obtain the rigidity in the third region R3, the thickness t3 of the tabular metallic base plate 109 is made larger than the thickness t2 in the second region R2. The thickness of the tabular metallic base plate 109 is increased discretely for the region where the components that cannot withstand a large load, such as the ICs 112, are disposed. Accordingly, a required rigidity can be obtained, while the weight of the tabular metallic base plate 109 can be controlled.

(39) In general, the ICs 112 tend to generate heat. The ICs 112 are mounted on the flexible circuit boards 111, which are typically thin and poor in thermal conductivity. Accordingly, the ICs 112 themselves can produce noises due to heat generation. Thus, a heat conductive rubber is used for the elastic member 120, which facilitates heat conduction to the tabular metallic base plate 109. An increase in the thickness t3 in the third region R3 improves heat conduction inside the tabular metallic base plate 109, which can reduce noise generation caused by the heat of the ICs 112.

(40) In the present embodiment, the thickness t3 in the third region R3 is made larger than the thickness t2 in the second region R2. The thickness t3 in the third region R3, however, can be equal to the thickness t2 in the second region R2.

(41) The fourth region R4 is a region where the battery holding member 113 is disposed. The battery holding member 113 accommodates and secures the battery 114. The battery 114 is armored by a resin cover that covers a storage unit and a circuit unit of the battery 114. The battery 114 itself can withstand a load. The battery 114 is secured so it contacts the tabular metallic base plate 109. More specifically, the battery 114 can always contact the tabular metallic base plate 109 or can contact the tabular metallic base plate 109 when the tabular metallic base plate 109 deforms. The gap between the tabular metallic base plate 109 and the battery 114 can be as small as possible.

(42) The thickness t4 of the fourth region R4 is made smaller than the basic thickness t1.

(43) Referring to FIGS. 5A and 5B, the following describes how the X-ray imaging apparatus provides a sufficient durability in the fourth region R4 even if the thickness t4 in the fourth region R4 is reduced. FIGS. 5A and 5B are views schematically illustrating a state where a load F is applied onto the X-ray imaging apparatus when the X-ray imaging apparatus is placed on a bed 121 with the front surface facing upward and a subject lies over the X-ray imaging apparatus.

(44) FIG. 5A illustrates a state where the load F is applied to the X-ray imaging apparatus while the battery 114 is not installed. FIG. 5B illustrates a state where the load F is applied to the X-ray imaging apparatus while the battery 114 is installed. The load F causes the entrance face plate 102 and the shock absorbing member 105 to deform. The deformation of the shock absorbing member 105 applies an additional load to the fluorescent member 106, the sensor panel 107, the shield member 108, and the tabular metallic base plate 109. In this case, if the battery 114 is not present as illustrated in FIG. 5A, the tabular metallic base plate 109 can deform at the battery accommodation space 113a. If the battery 114 is present as illustrated in FIG. 5B, the battery 114 and the tabular metallic base plate 109 areally contact each other, which prevents local deformation of the tabular metallic base plate 109 and thereby reduces the deformation of the tabular metallic base plate 109. The thickness t4 in the fourth region R4 can then be made smaller than the basic thickness t1, which can reduce the weight of the apparatus.

(45) In general, the capacity of the battery 114 increases in proportion to the size thereof. The size of the battery 114 can increase as the thickness t4 in the fourth region R4 decreases, which advantageously leads to an increase in the capacity of the battery 114.

(46) As described above, the rear surface of the tabular metallic base plate **109** are divided into multiple regions depending on members disposed therein, and the thicknesses and the disposal of the ribs **109a** can be set appropriately in respective regions to obtain an appropriate balance between weight reduction and increased strength.

(47) In the X-ray imaging apparatus of the present embodiment, as illustrated in FIG. 3B, the tabular metallic base plate **109** is shaped such that the thickness gradually decreases in at least one in-plane direction (in the downward direction in FIG. 3B). The shape where the thickness gradually decrease in an in-plane direction is advantageous in manufacturing the tabular metallic base plate **109** because in metal forming, such as die casting or thixotropic casting, the material can flow easily from the thick side of the plate.

(48) The basic thickness t_1 of the tabular metallic base plate **109** can be approximately 1.5 mm, which enables metal forming relatively reliably while suppressing weight increase. A required strength can be obtained by the location of the ribs **109a**. Accordingly, an appropriate balance between weight reduction and increased strength can be obtained.

(49) The thickness in the second region **R2** where the electric circuit boards **110** are disposed can be 2.0 mm or more. When the size of the tabular metallic base plate **109** is approximately 12,000 mm^{sup.2}, an approximate thickness of 2.0 mm in the second region **R2** can provide an appropriate strength without increasing the mass excessively, which leads to an appropriate balance between weight reduction and increased strength.

(50) The thickness in the fourth region **R4** where the battery **114** is disposed can be less than 1.5 mm. An approximate thickness of 1.2 mm in the fourth region **R4** can provide an appropriate strength when the battery **114** is present. This thickness is feasible in the metal forming if this thickness is local. This can reduce the mass of the X-ray imaging apparatus.

(51) The present disclosure has been described through the embodiment, which is merely an example for implementing the present disclosure and accordingly should not be construed as limiting the technical scope of the present disclosure. In other words, the present disclosure can be implemented in various other forms without departing from the technical idea and the main features of the disclosure.

(52) According to the present disclosure, an appropriate balance between weight reduction and increased strength can be obtained for the X-ray imaging apparatus.

(53) While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(54) This application claims the benefit of Japanese Patent Application No. 2021-173697 filed Oct. 25, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. A radiographing apparatus, comprising: a sensor panel configured to convert incident radiation to an electric signal; and a tabular metallic base plate supporting the sensor panel, the tabular metallic base plate including a plurality of flat regions among which at least three flat regions are arranged in a predetermined direction, wherein the thickness of the plurality of flat regions decreases sequentially in the predetermined direction.

2. The radiographing apparatus according to claim 1, wherein the tabular metallic base plate is formed by die casting or thixomolding.

3. The radiographing apparatus according to claim 1, wherein a first flat region among the plurality of flat regions is a region where no circuit board is provided, a second flat region among the plurality of flat regions is a region where a circuit board is provided, and the second flat region is thicker than the first flat region.

4. The radiographing apparatus according to claim 1, wherein a second flat region among the plurality of flat regions is a region where a circuit board is provided, a third flat region among the plurality of flat regions is a region facing an IC chip configured to amplify the electric signal from the sensor panel, and the third flat region is thicker than the second flat region.
5. The radiographing apparatus according to claim 1, wherein a fourth flat region among the plurality of flat regions is a region with which a power supply member is in contact, a first flat region among the plurality of flat regions is a region in which the power supply member is not in contact, and the fourth flat region is thinner than the first flat region.
6. The radiographing apparatus according to claim 5, further comprising: a housing configured to accommodate the sensor panel and the tabular metallic base plate, the housing including a holding portion for holding the power supply member, wherein the power supply member and the holding portion are in contact with the fourth flat region.
7. The radiographing apparatus according to claim 5, wherein the thickness of the fourth flat region is less than 1.5 mm.
8. The radiographing apparatus according to claim 5, wherein the fourth flat region extends to at least one end of the tabular metallic base plate.
9. The radiographing apparatus according to claim 5, wherein the power supply member is a battery.
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